Simulation Arena

Deal Approval & Margin Balancing

Approve 0% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $3,835.12M of expected profit.

Seed 42 | Run run_a2128b9dee98 | 170,000 scenarios tested | Monte Carlo simulation with an empirica… | 21 Sep 2026, 20:56

Same seed, same numbers - this run can be reproduced exactly.

How this works

For each discount level we ran thousands of 'what-if' worlds. In each one the buyer's odds of saying yes wobble a little, and your delivery cost wobbles a little. We then ask the only question that matters: if you offered this discount again and again, how much profit would you bank on average? That is the expected profit - the chance of winning multiplied by the profit you keep. Because those worlds are a sample, we also work out how much the answer itself would wobble if we ran it again. Any discount whose expected profit is within twice that wobble of the best - or within 1% of the best profit, whichever is wider - is treated as giving the same answer. That is the range where you have genuine room to negotiate.

The numbers that matter

BEST DISCOUNT TO OFFER
0%
This is where the extra wins are still worth more than the margin given away.
vs rep asked for 25%

EXPECTED PROFIT AT THAT DISCOUNT
$489.79M
Average profit if you made this call many times over.

DISCOUNT RANGE YOU CAN APPROVE WITHOUT LOSING MONEY
0%
Only one discount stands out - the model cannot tell any neighbouring discount apart from it. The model's own margin of error here is about $38.00M, so treat this as the best single number - the next discount along already costs real money.
vs rep asked for 25% - outside the range

WHERE IT STARTS TO GENUINELY COST YOU
2%
At 2% the expected profit falls $191.78M below the best. That gap is wider than the model's own margin of error, so the loss is real.
vs costs about $191.78M

EXPECTED PROFIT IF YOU CAVE IN
-$3,345.34M
What 25% actually earns you.
vs -$3,835.12M vs the optimum

PROFIT GIVEN AWAY
$3,835.12M
Buying +49 points of win probability costs you this much.

ODDS OF WINNING AT THE BEST DISCOUNT
36.9%
The chance the deal lands.

WORST CASE (1-IN-20)
$0
In an unlucky run of 20 deals, this is roughly the profit you would see.

How we got there

1. The two forces fighting each other

Every point of discount does two things: it makes the buyer a bit more likely to say yes, and it takes $209.12M straight off your profit. Early on the first effect wins. Later, the second one does.

Evidence: Cost to deliver $19,593.53M on $20,911.98M of list value

2. What the buyer's behaviour looks like

They start around 35.0% likely to buy at full price and never become certain - the curve flattens near 95.0%. Around 17% discount is where they are most persuadable.

Evidence: Curve source: 20000 of your past deals

3. The sweet spot

0% discount gives the highest expected profit at $489.79M, with a 36.9% chance of winning.

Evidence: peak of 17 levels tested

4. The range where the answer is the same

0% is the only discount that stands out; the model cannot tell any neighbouring discount apart from it. The next level along already costs $191.78M.

Evidence: margin of error $38.00M; safe range 0%

5. What the requested discount really costs

Moving from 0% to 25% buys +49 points of win probability but costs $3,835.12M of expected profit - more than the model's margin of error, so the loss is real.

Evidence: -$3,345.34M vs $489.79M

Charts

Where profit peaks

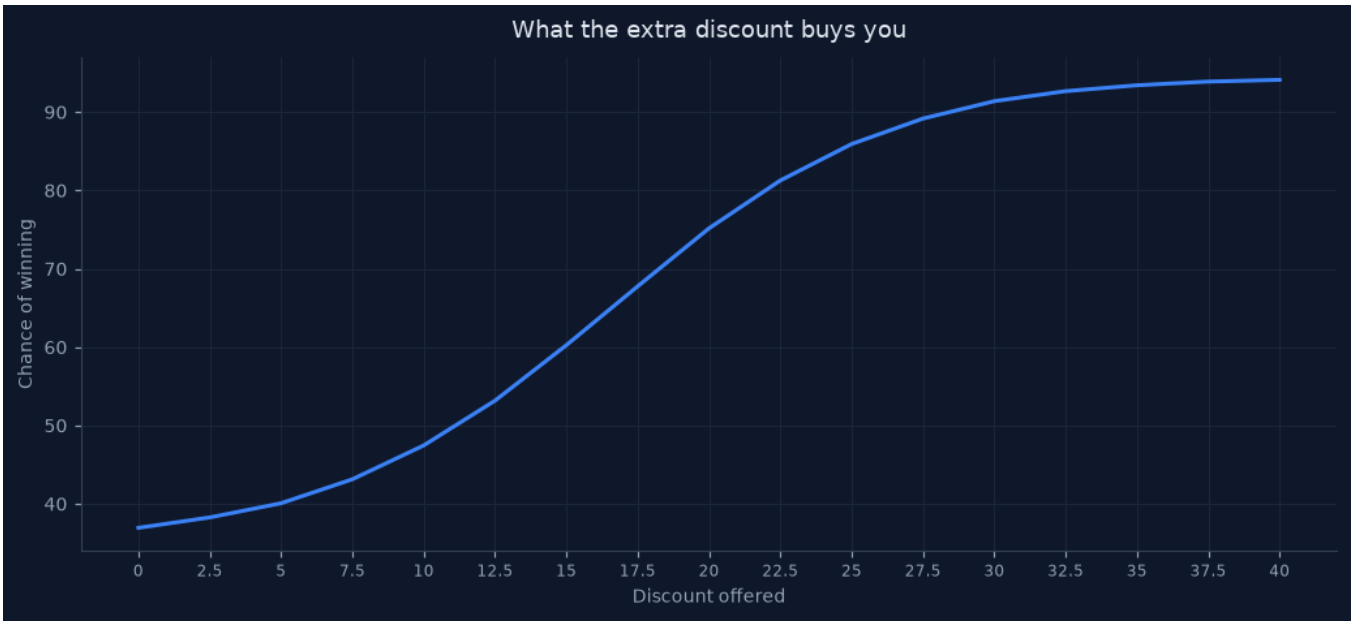
Everything from 0% gives the same answer within the model's margin of error; the peak is 0%



What this shows: Grey is what you keep per won deal - it only falls. Green is what you actually expect to earn - it rises, peaks, then falls. The amber dashed line is the best profit minus the model's own margin of error: every discount whose green line stays above it is the same answer, which is the 0% range. Past that you are genuinely paying for the win.

What the extra discount buys you

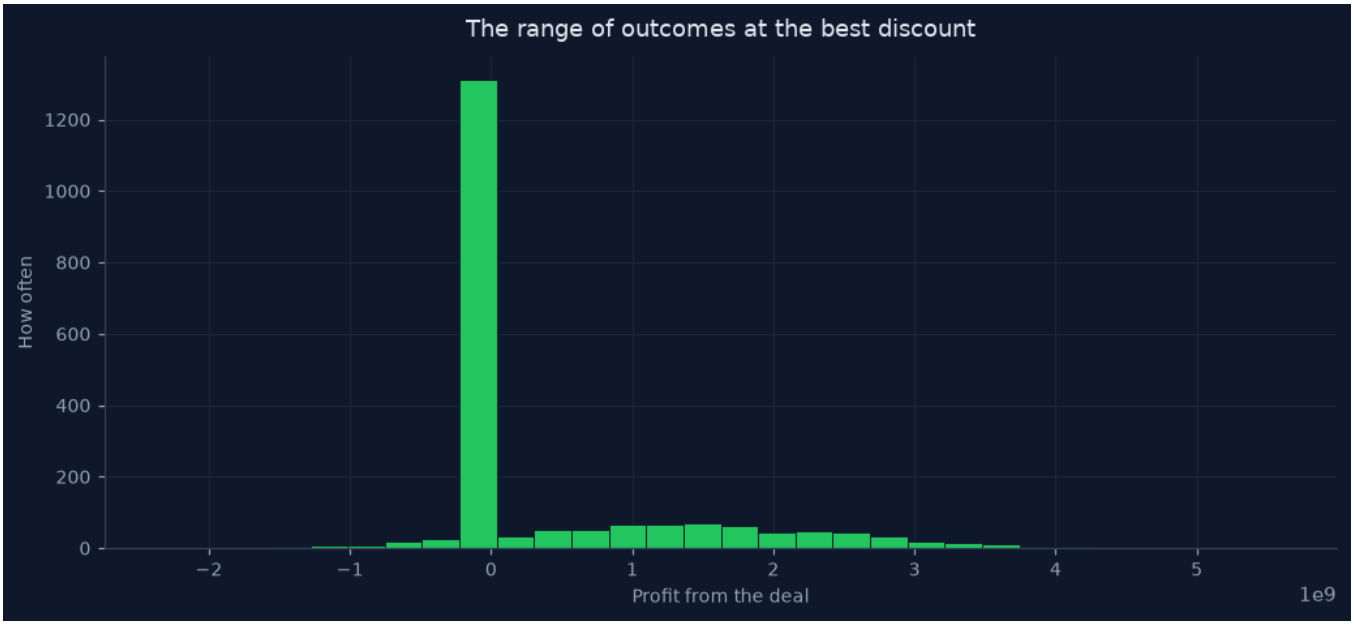
Curve learned from 20000 of your past deals



What this shows: Past the knee of this curve, extra discount buys almost no extra certainty - but keeps costing margin. That is the trap.

The range of outcomes at the best discount

10,000 simulated worlds at 0% discount



What this shows: The cluster at zero is the deals you lose. The spread on the right is the profit you make when you win.

Tables

Every discount level tested

Expected profit = chance of winning x profit kept. Highlighted row is the winner.

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
0.0%	$20,911,979,419	$1,316,300,252	36.9%	$489,788,040	$0	85% chance
2.5%	$20,389,179,933	$806,594,813	38.3%	$298,003,747	-$518,160,848	92% chance
5.0%	$19,866,380,448	$271,925,292	40.1%	$107,035,517	-$1,106,043,708	96% chance
7.5%	$19,343,580,962	-$257,233,422	43.1%	-$108,229,562	-$1,667,770,697	99% chance
10.0%	$18,820,781,477	-$777,711,373	47.5%	-$359,508,475	-$2,164,567,940	100% chance
12.5%	$18,297,981,991	-$1,260,773,237	53.2%	-$665,850,777	-$2,766,028,834	100% chance

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
15.0%	$17,775,182,506	-$1,811,949,781	60.3%	-$1,098,691,257	-$3,453,761,503	100% chance
17.5%	$17,252,383,021	-$2,312,040,617	67.8%	-$1,562,362,933	-$4,013,868,439	100% chance
20.0%	$16,729,583,535	-$2,849,981,803	75.2%	-$2,143,707,264	-$4,636,785,211	100% chance
22.5%	$16,206,784,050	-$3,382,134,825	81.3%	-$2,759,835,270	-$5,202,826,247	100% chance
25.0%	$15,683,984,564	-$3,911,588,991	85.9%	-$3,345,336,695	-$5,754,713,952	100% chance
27.5%	$15,161,185,079	-$4,442,147,949	89.2%	-$3,950,806,161	-$6,295,583,075	100% chance
30.0%	$14,638,385,593	-$4,982,764,419	91.4%	-$4,556,191,989	-$6,898,065,080	100% chance
32.5%	$14,115,586,108	-$5,484,029,348	92.7%	-$5,082,629,649	-$7,407,158,713	100% chance
35.0%	$13,592,786,622	-$6,019,753,538	93.4%	-$5,631,830,290	-$7,901,217,889	100% chance
37.5%	$13,069,987,137	-$6,505,024,880	93.9%	-$6,104,155,757	-$8,375,361,830	100% chance
40.0%	$12,547,187,651	-$7,047,045,555	94.1%	-$6,637,161,913	-$8,937,063,215	100% chance

Row 1 maximises expected profit.

The decision in one line

What changes if you approve the discount being asked for.

Column 1	Best (0%)	Asked (25%)	Difference
Discount	0.0%	25.0%	+25.0 pts
Win chance	36.9%	85.9%	+49.0 pts
Profit if won	$1,316,300,252	-$3,911,588,991	-$5,227,889,243
Expected profit	$489,788,040	-$3,345,336,695	-$3,835,124,735
Range where the answer is the same	0%	outside it	margin of error $38.00M

The range where the answer is the same

Every discount we tested, how far it falls short of the best, and whether that shortfall is real money or just the model's own margin of error.

Discount	Expected profit	Short of the best	Inside the range?	What it means
0.0%	$489,788,040	$0	Inside	safe to approve
2.5%	$298,003,747	$191,784,293	Outside	costs you $191.78M
5.0%	$107,035,517	$382,752,523	Outside	costs you $382.75M
7.5%	-$108,229,562	$598,017,602	Outside	costs you $598.02M
10.0%	-$359,508,475	$849,296,515	Outside	costs you $849.30M
12.5%	-$665,850,777	$1,155,638,816	Outside	costs you $1,155.64M
15.0%	-$1,098,691,257	$1,588,479,296	Outside	costs you $1,588.48M
17.5%	-$1,562,362,933	$2,052,150,972	Outside	costs you $2,052.15M
20.0%	-$2,143,707,264	$2,633,495,304	Outside	costs you $2,633.50M
22.5%	-$2,759,835,270	$3,249,623,310	Outside	costs you $3,249.62M
25.0%	-$3,345,336,695	$3,835,124,735	Outside	costs you $3,835.12M
27.5%	-$3,950,806,161	$4,440,594,201	Outside	costs you $4,440.59M
30.0%	-$4,556,191,989	$5,045,980,028	Outside	costs you $5,045.98M
32.5%	-$5,082,629,649	$5,572,417,689	Outside	costs you $5,572.42M
35.0%	-$5,631,830,290	$6,121,618,330	Outside	costs you $6,121.62M
37.5%	-$6,104,155,757	$6,593,943,796	Outside	costs you $6,593.94M
40.0%	-$6,637,161,913	$7,126,949,953	Outside	costs you $7,126.95M

Highlighted rows are the range where the answer is the same (0%) - within about $38.00M of the best.

Recommended discount per deal (top 25)

The same logic applied to each deal individually.

Deal	Customer	Value	Margin	Best discount	Expected profit
Q-000011589	Acct-226235	$24,000,000	-5.7%	0%	-$502,859
Q-000009396	Acct-223884	$24,000,000	3.8%	0%	$337,013
Q-000009047	Acct-090640	$24,000,000	4.8%	0%	$424,814
Q-000008091	Acct-110274	$19,296,863	15.4%	0%	$1,096,718
Q-000007533	Acct-062342	$16,052,463	0.7%	0%	$40,337
Q-000018826	Acct-138872	$14,565,883	23.4%	0%	$1,260,595
Q-000002411	Acct-019732	$14,340,308	-11.8%	0%	-$623,716
Q-000006273	Acct-125458	$14,145,541	46.0%	20%	$2,766,932
Q-000005246	Acct-233590	$14,094,974	12.6%	0%	$654,193
Q-000013379	Acct-205409	$13,878,192	21.0%	0%	$1,078,510
Q-000004222	Acct-219531	$13,822,773	2.4%	0%	$123,102
Q-000009736	Acct-113136	$13,452,114	-3.5%	0%	-$172,493
Q-000017300	Acct-007220	$13,422,614	25.1%	0%	$1,245,477
Q-000000139	Acct-108061	$13,011,806	-15.3%	0%	-$736,628
Q-000009880	Acct-201403	$12,782,137	-22.3%	0%	-$1,053,320
Q-000001755	Acct-058430	$12,706,362	-8.4%	0%	-$393,945
Q-000001411	Acct-122168	$12,567,749	23.2%	0%	$1,077,915
Q-000010633	Acct-176419	$12,154,931	1.3%	0%	$59,290
Q-000013798	Acct-106373	$11,654,026	18.1%	0%	$779,052
Q-000010069	Acct-070297	$11,489,319	15.7%	0%	$665,721
Q-000013273	Acct-130561	$11,410,414	27.0%	0%	$1,136,772
Q-000007138	Acct-039861	$11,334,211	18.2%	0%	$763,956
Q-000005723	Acct-178920	$11,250,338	16.4%	0%	$680,975
Q-000001338	Acct-100033	$11,225,300	17.8%	0%	$739,607
Q-000009264	Acct-188657	$11,173,578	22.0%	0%	$908,792

What we recommend

Approve 0% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $3,835.12M of expected profit.

Risk High 100 out of 100 | Confidence: High

Why

• Only 0% stands out from the rest - the model cannot tell any neighbouring discount apart from it.

• Expected profit peaks at 0% ($489.79M).

• The requested 25% delivers -$3,345.34M - $3,835.12M less than the peak, which is more than the model's margin of error.

• Win probability only improves +49 points for that giveaway.

What to do

1. Approve 0% as the standard concession - it is the only discount the model can tell apart from the rest.

2. Above 0% (the top of the safe range), require VP sign-off: the 25% being asked for costs about $153.40M per point of discount.

3. At 25% there is a 100% chance of breaching the 12.0% margin floor - add a volume or term commitment to earn the deeper discount.

The trade-off: You are trading $3,835.12M of expected profit for +49 points of win probability.

What we assumed and what to watch

Worth knowing before you act

- At 25% discount there is a 100% chance the deal actually costs you money to deliver even if you win it.

Assumptions

- List value $20,911,979,419, cost $19,593,528,342 (9.4% standard margin).
- Consumption effects: 0.2% of value is consumed as unmetered burst you still pay for; 4 months of rollover defers revenue.
- Win probability curve: from 35.0% at full price toward a ceiling of 95.0%, halfway at 16.6% discount (source: 20000 of your past deals).
- Win probability uncertainty: plus or minus 5 points.
- Cost volatility: plus or minus 6% of cost.
- Margin floor: 12.0%.
- Two discounts are treated as giving the same answer when the gap between their expected profits is smaller than twice the model's own margin of error at both discounts added together (about $38.00M at the best one), or smaller than 1% of the best profit - whichever is larger.
- The range where the answer is the same is 0%, taken from the 17 discount levels we tested.
- 10,000 simulations per discount level; seed 42 (re-running gives identical numbers).
- Deal / quote ID -> your 'deal_id' column
- Customer name -> your 'customer' column
- Deal value (list price) -> your 'deal_value' column
- Cost to deliver -> your 'cost' column
- Standard margin (%) -> your 'margin_pct' column
- Discount given (%) -> your 'discount_pct' column
- Won? (1/0) -> your 'won' column
- Hit a consumption tier accelerator -> your 'tier_accelerator_hit' column
- Unmetered burst used -> your 'burst_allowance_pct' column
- Rollover window -> your 'rollover_window_months' column

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
0.00s	prepare	Won? (1/0) -> your 'won' column
0.00s	prepare	Hit a consumption tier accelerator -> your 'tier_accelerator_hit' column
0.00s	prepare	Unmetered burst used -> your 'burst_allowance_pct' column
0.00s	prepare	Rollover window -> your 'rollover_window_months' column
0.77s	prepare	Read 20,000 records into the model.
0.77s	simulate	Running the simulation…
0.79s	read	Reading the deal you gave us...
0.81s	read	Found 20000 deal(s) worth $20,911.98M at an average 9.4% margin.
0.81s	prepare	Cost to deliver is about $18,937.41M.

Time	Stage	What happened
0.83s	prepare	Added the consumption effects from your file: $656.12M of extra cost to serve.
0.83s	prepare	Working out how much a discount really moves the buyer...
1.20s	prepare	Learned the buyer's behaviour from 20000 past deals (they flip hardest around 16.6% discount).
1.20s	simulate	Testing 17 discount levels, 10,000 outcomes each (170,000 scenarios in total)...
1.22s	simulate	Simulated 11% of the discount levels...
1.22s	simulate	Simulated 23% of the discount levels...
1.22s	simulate	Simulated 35% of the discount levels...
1.22s	simulate	Simulated 47% of the discount levels...
1.22s	simulate	Simulated 58% of the discount levels...
1.22s	simulate	Simulated 70% of the discount levels...
1.23s	simulate	Simulated 82% of the discount levels...
1.23s	simulate	Simulated 94% of the discount levels...
1.23s	simulate	Simulated 100% of the discount levels...
1.23s	summarize	All discount levels tested. Looking for the sweet spot...
1.23s	summarize	Checked how much the answer would wobble if we ran it again - the range where the answer is the same is 0%.
1.26s	summarize	Done. Best discount is 0% ($489.79M expected profit); the range where the answer is the same is 0%.